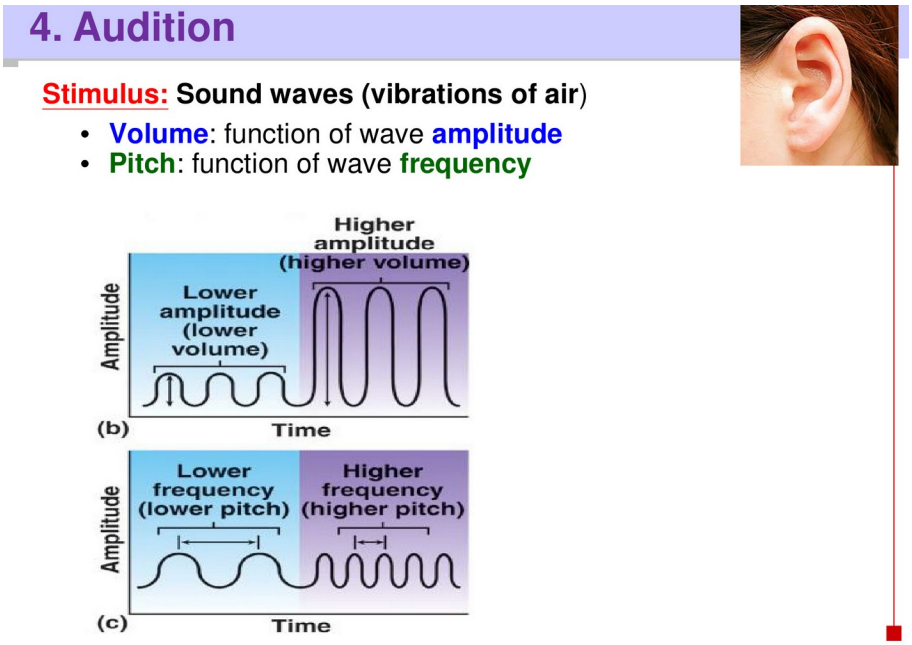
01 Hearing & balance: two special senses

Both use inner-ear hair cells (mechanoreceptors)
→ cranial nerve VIII (vestibulocochlear nerve) → brainstem.

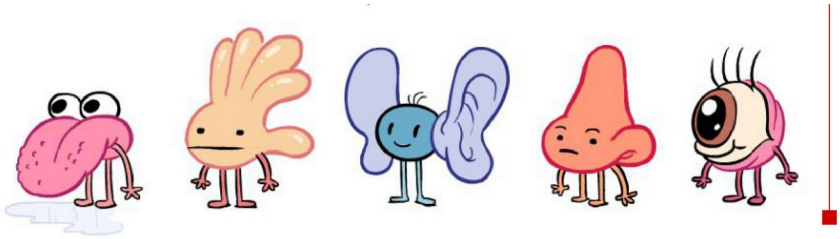
System	Stimulus	Receptor region / nerve branch
Audition	Sound waves: vibrations of air.	Inner hair cells in the cochlear Organ of Corti → cochlear / auditory branch of VIII.
Vestibular sensation	Head movement and orientation relative to gravity.	Hair cells in otolith organs and semicircular canals → vestibular branch of VIII.

Sound: amplitude and frequency carry different information



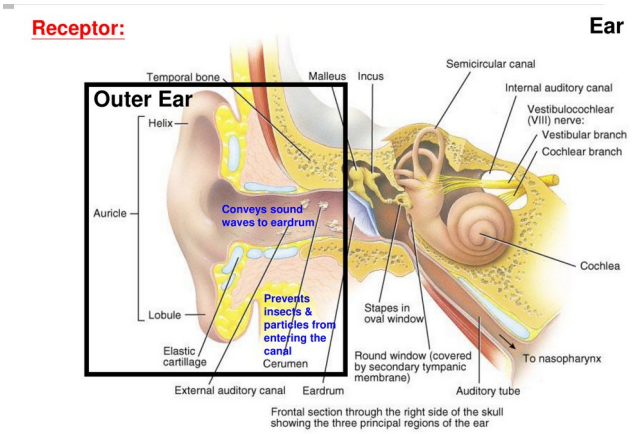
Wave property	Perceptual association
Amplitude	Greater amplitude → louder sound / greater perceived volume.
Frequency	More oscillations per second → higher pitch. Fewer oscillations per second → lower pitch.

- Changing loudness does not necessarily change pitch: amplitude and frequency are separate wave properties.

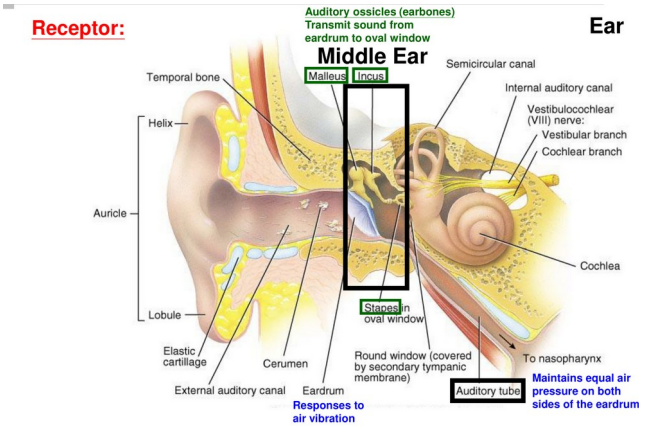


02 Outer & middle ear: deliver vibrations

Sound waves → **external auditory canal → tympanic membrane (eardrum) → malleus → incus → stapes → oval window**



Outer ear collects and conducts sound.

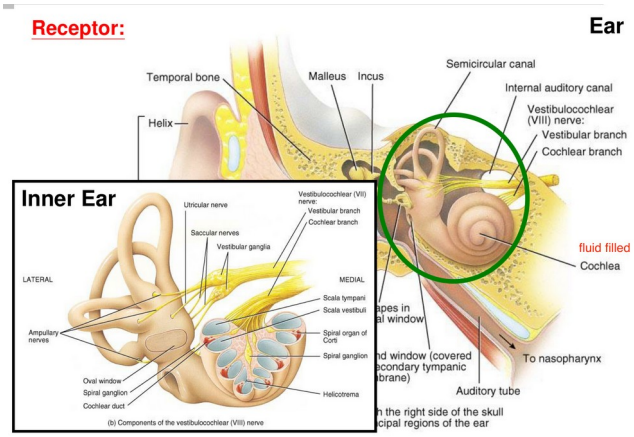


Middle-ear ossicles transmit the vibration.

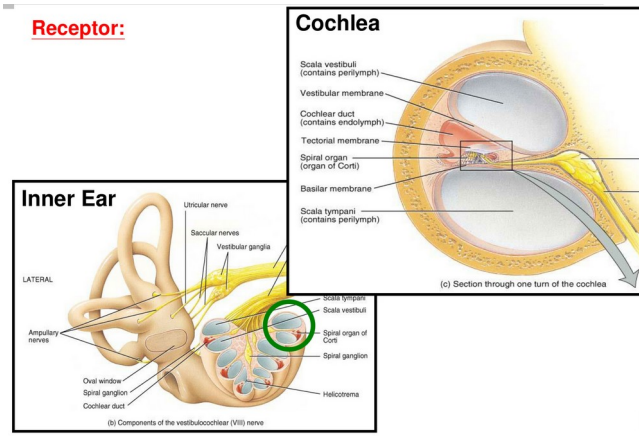
Structure	Function
Auricle / pinna	Collects sound at the entrance to the outer ear.
External auditory canal	Conveys sound waves to the eardrum.
Cerumen (earwax)	Helps prevent insects and particles from entering farther into the canal.
Tympanic membrane	Vibrates in response to changes in air pressure.
Auditory ossicles	Malleus, incus and stapes → mechanical transmission from eardrum to oval window.
Auditory / Eustachian tube	Connects the middle ear with the nasopharynx → helps equalise air pressure on both sides of the eardrum.
Oval vs round window	Stapes acts at the oval window. The flexible round-window membrane accommodates pressure changes in the cochlear fluid.

- **Eardrum → three ossicles → oval window** links an airborne stimulus to the fluid-filled inner ear.

03 Inner ear & the three cochlear compartments



The inner ear is fluid-filled.



The cochlear cross-section separates three spaces.

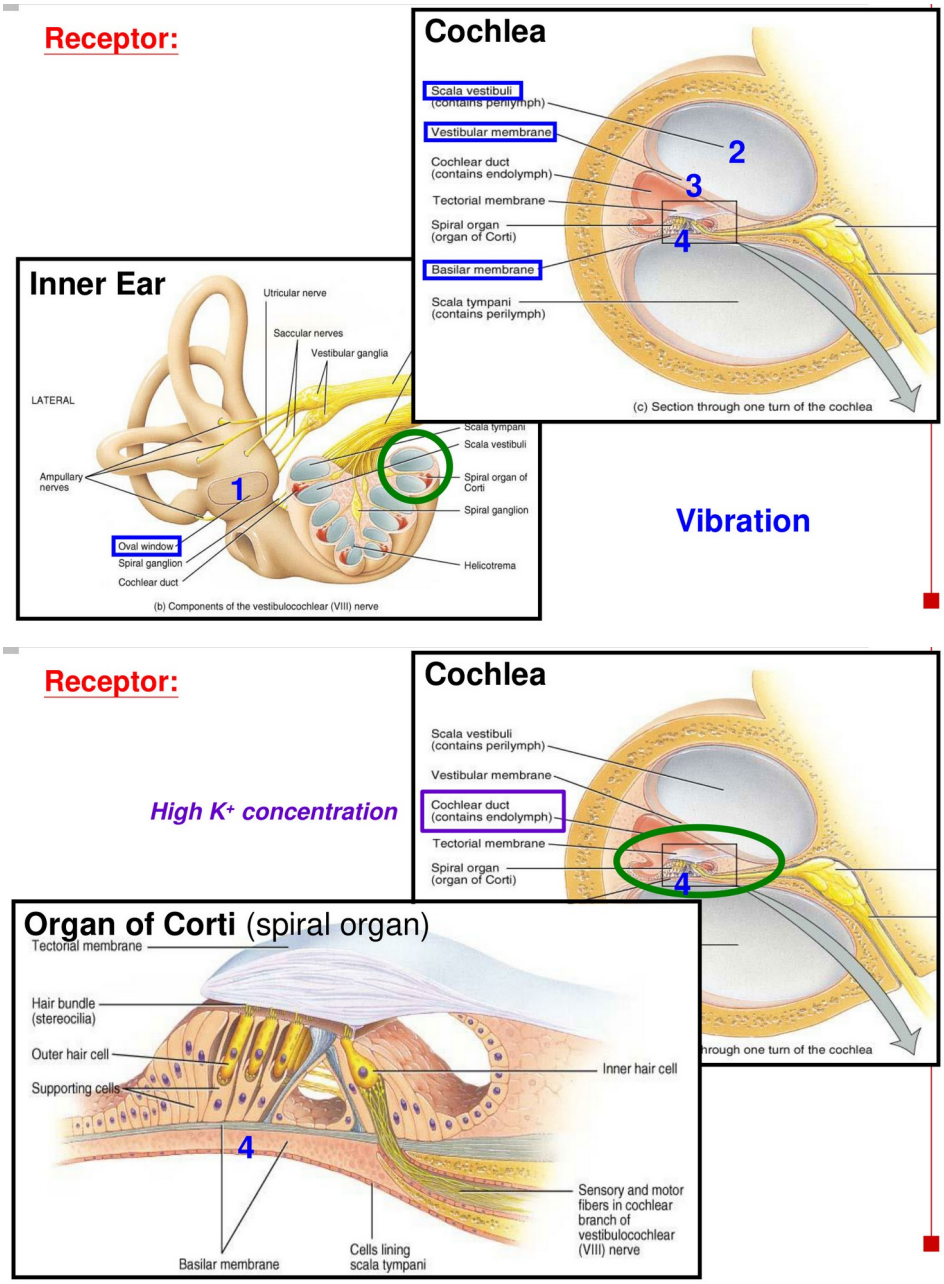
Compartment / membrane	Location and role
Scala vestibuli	Upper cochlear compartment; contains perilymph and receives pressure changes transmitted at the oval window.
Cochlear duct / scala media	Middle compartment; contains **K^+-rich endolymph**.
Scala tympani	Lower compartment; contains perilymph and communicates mechanically with the round window.
Vestibular membrane	Separates scala vestibuli from the cochlear duct.
Basilar membrane	Separates the cochlear duct from scala tympani; supports the Organ of Corti.
Tectorial membrane	Overlies the hair-cell apparatus; relative motion helps bend hair bundles.
Spiral ganglion / cochlear nerve	Sensory afferent cell bodies and fibres carry cochlear output toward the brainstem.

Perilymph: scala vestibuli + scala tympani.
Endolymph: cochlear duct → high extracellular K^+ around the hair bundles.

- The cochlea serves hearing; the neighbouring utricle, saccule and semicircular canals serve vestibular sensation.

04 From oval window to the Organ of Corti

Stapes vibrates at the **oval window** → cochlear fluid-pressure waves
→ displacement of the cochlear partition / **basilar membrane**
→ movement of the **Organ of Corti** → hair-bundle deflection



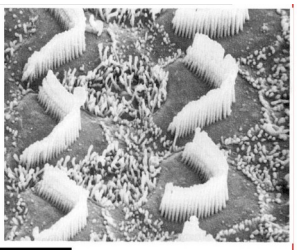
- **Organ of Corti = spiral organ**, the sensory structure of the cochlea.
- The basilar membrane carries the hair cells; the overlying tectorial membrane moves differently → relative shear stimulates the bundles.
- Pressure waves transfer mechanical energy. Endolymph and perilymph remain in their respective compartments; normal transduction does not require them to mix.

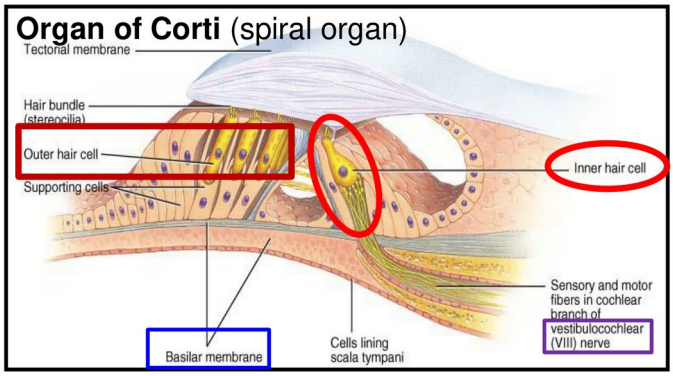
05 Inner vs outer hair cells

Receptor cell:

Inner hair cells (mechanoreceptors)
Stimulated by vibration of the basilar membrane

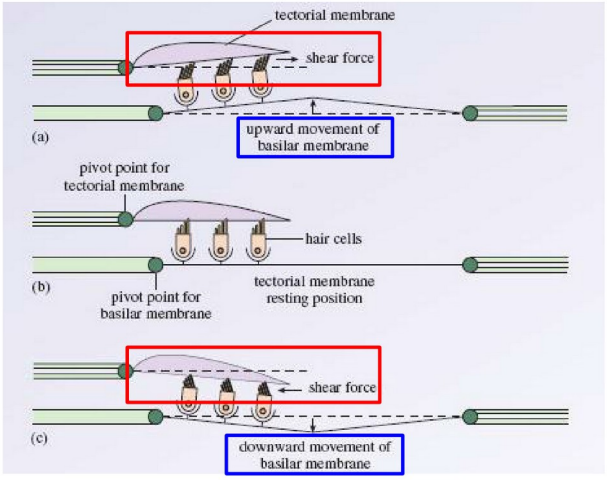
Outer hair cells
Develop electromotility due to prestin contraction





Cell type	Main role
Inner hair cells (IHCs)	Principal sensory receptors for hearing. Basilar-membrane vibration bends the bundle → receptor potential → transmitter release to auditory afferents.
Outer hair cells (OHCs)	Develop **electromotility through prestin**: voltage-dependent changes in cell length feed mechanical energy back into cochlear motion, enhancing sensitivity and frequency tuning.

Displacement of the basilar membrane creates shearing forces between hair cells & tectorial membrane.



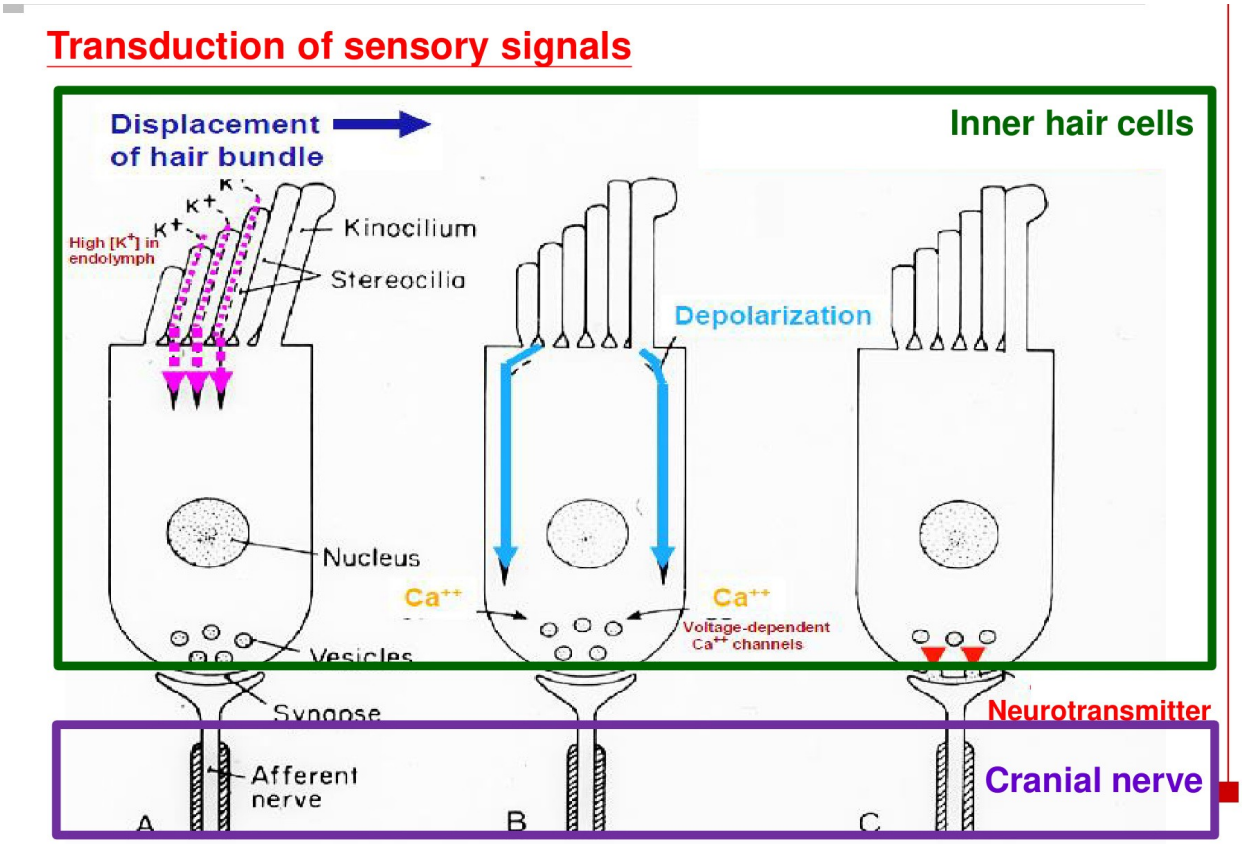
No stimulation

Basilar membrane moves up / down
→ **shearing forces between the hair-cell apparatus and tectorial membrane**
→ stereocilia deflect → hair-cell electrical activity changes

- The resting position lacks the sound-driven displacement shown by the upward and downward phases; opposite deflections have opposite electrical effects.
- **IHCs chiefly encode the sound; OHCs chiefly modify and amplify the mechanics.**

06 Hair-cell transduction: K^+ first, Ca^{2+} next

Excitatory displacement of the **stereocilia / hair bundle**
→ mechanically gated transduction channels open
→ **K^+ influx from endolymph → hair-cell depolarisation**
→ **voltage-gated Ca^{2+} channels open → Ca^{2+} influx**
→ **neurotransmitter release ↑ → auditory afferent firing ↑**



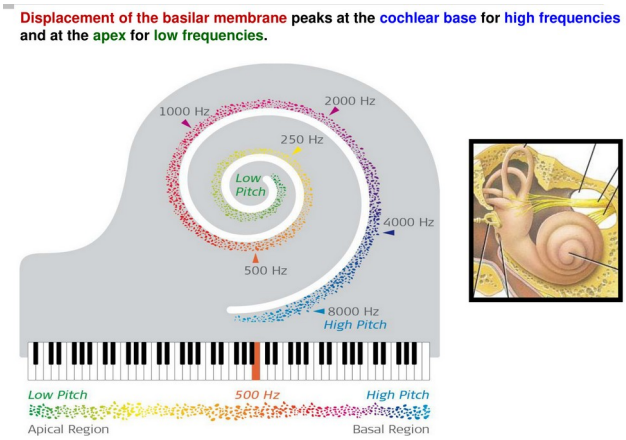
Step	Why it matters
K^+ entry at the hair bundle	Endolymph is rich in K^+; the electrochemical gradient favours K^+ entry into the hair cell. Inward positive charge depolarises it.
Ca^{2+} entry near the synapse	Depolarisation opens voltage-gated Ca^{2+} channels; Ca^{2+} triggers transmitter release.
Hair cell vs nerve fibre	The hair cell produces a graded receptor potential. The afferent nerve produces action potentials.

- An inhibitory bend closes more transduction channels → less inward current → hyperpolarisation and reduced transmitter release.

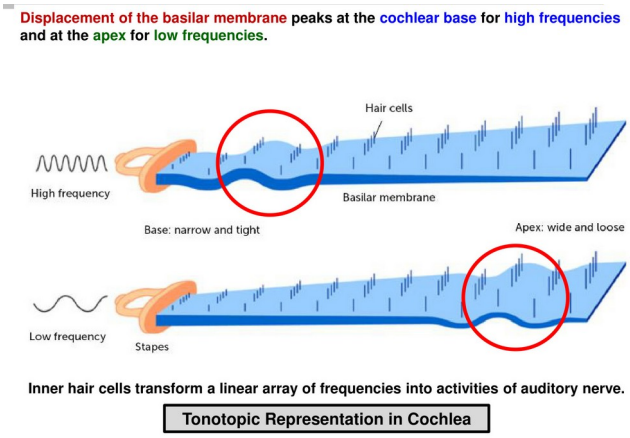
In mature cochlear hair cells, excitatory bending is toward the tallest stereocilia; vestibular hair cells retain a kinocilium that provides the directional reference.

07 Cochlear tonotopy: frequency selects a place

Region	Basilar-membrane mechanics	Peak response
Base	**Narrow and stiff**; near the oval window.	**High frequencies**
Apex	**Wide and floppy / compliant**; toward the cochlear tip.	**Low frequencies**



A spatial frequency map follows the cochlear spiral.



High and low frequencies peak at different positions.

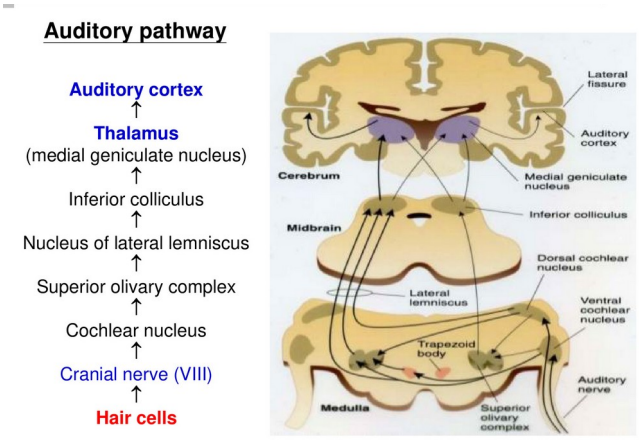
Sound frequency → location of maximum basilar-membrane displacement
→ **local inner hair cells are most strongly stimulated**
→ activity in corresponding auditory nerve fibres

- **Inner hair cells transform a linear array of frequencies into auditory nerve activity.**
- High-frequency sound peaks near the stiff base; lower-frequency waves reach their greatest displacement farther toward the compliant apex.
- The frequency map is continuous: intermediate frequencies peak at intermediate locations.

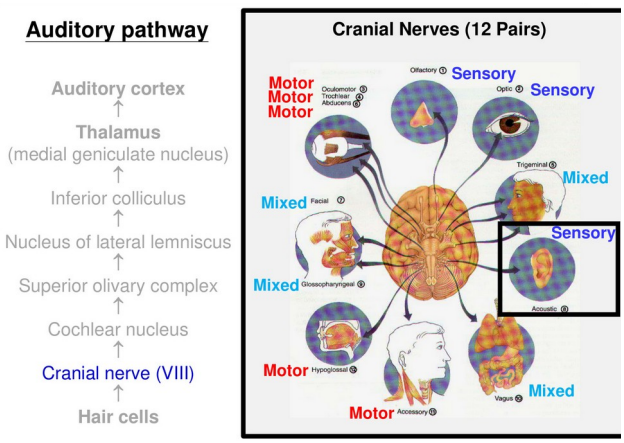
Base = high pitch. Apex = low pitch.
This is tonotopic organisation: frequency is mapped onto position.

08 Auditory pathway: cochlea to temporal cortex

Inner hair cells → cochlear branch of CN VIII → cochlear nuclei
→ superior olivary complex → lateral lemniscus / its nuclei
→ inferior colliculus → medial geniculate nucleus (MGN) of the thalamus
→ auditory cortex in the temporal lobe



Brainstem relays → thalamus → cortex



CN VIII carries sensory information.

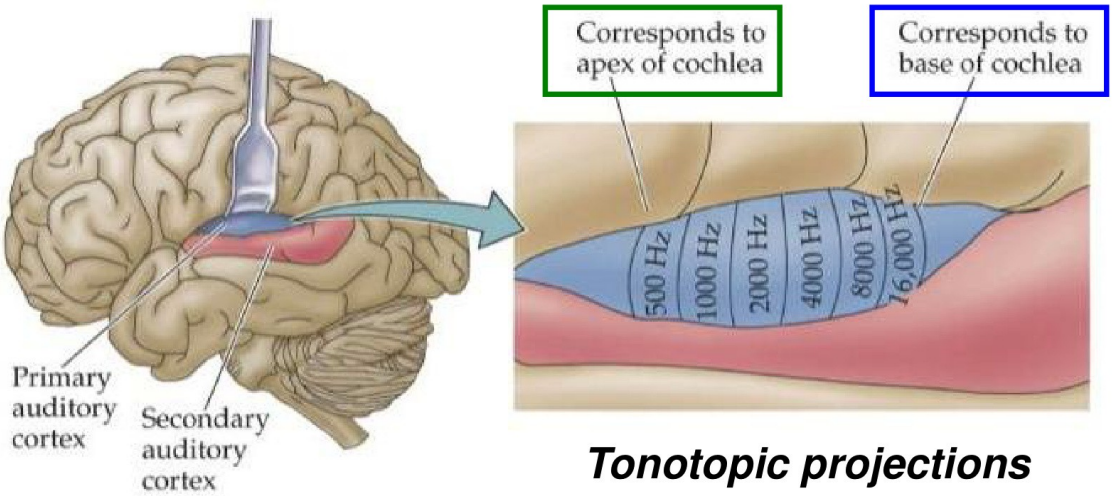
Relay / structure	Location or role
Cochlear nuclei	First central auditory relay in the brainstem; dorsal and ventral divisions.
Superior olivary complex	Compares information from the two ears.
Lateral lemniscus	Ascending auditory pathway with associated relay nuclei.
Inferior colliculus	Midbrain auditory relay.
MGN	Thalamic auditory relay → auditory cortex.
Auditory cortex	Temporal lobe; cortical processing of sound.

- Ascending auditory pathways have extensive **bilateral connections** after the cochlear nuclei; the diagram includes crossing fibres and the trapezoid body.

09 Tonotopy continues into the brain

Cochlear frequency map → auditory nerve activity
→ **similarly tuned neurons cluster together at successive relays**
→ organised frequency representation in the auditory cortex

Primary auditory cortex is arranged as functional columns.



Tonotopic Representation in Primary Auditory Cortex

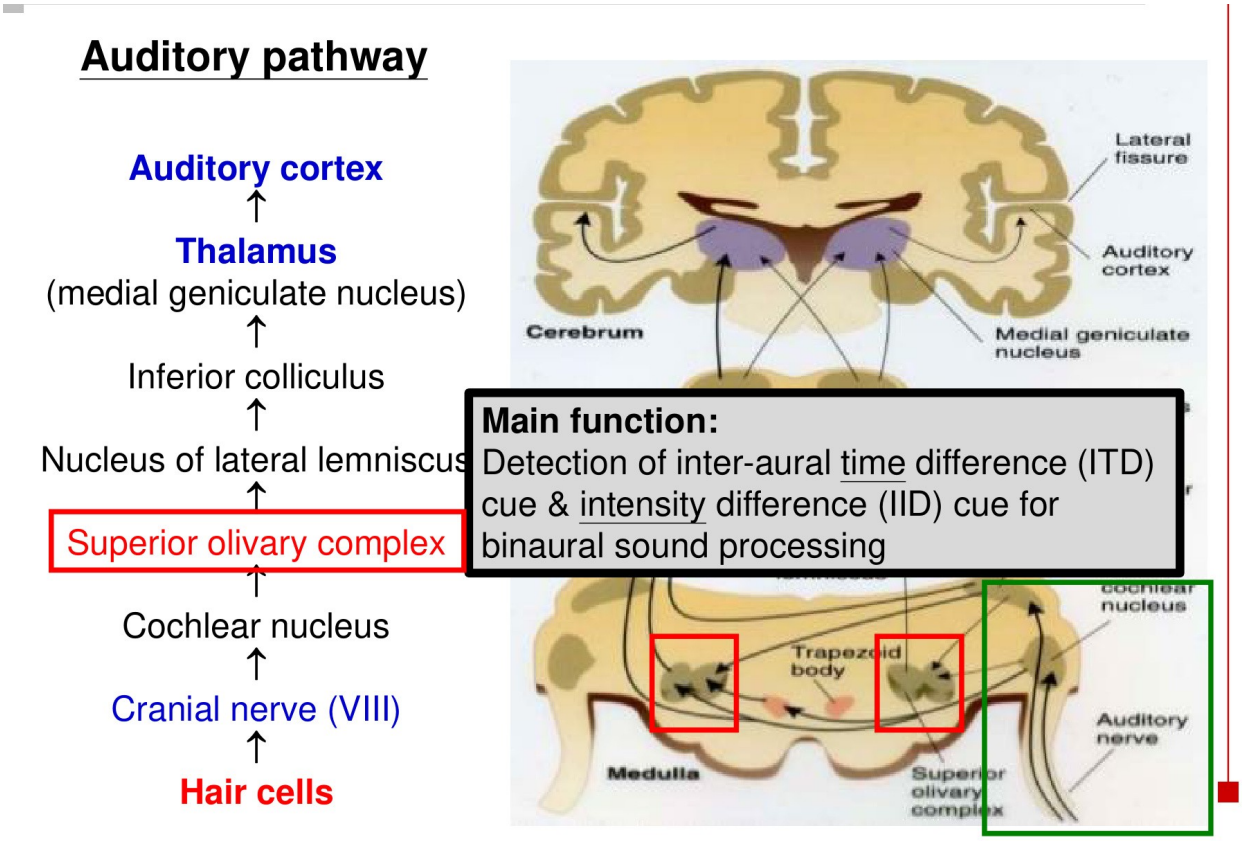
Level	Organising rule
Cochlea	Basilar-membrane position determines the frequency producing peak displacement.
Ascending auditory relays	Neurons sensitive to similar frequencies remain clustered together.
Primary auditory cortex	Arranged as **functional columns according to a sound-frequency map**.

- Regions representing the **cochlear apex** process relatively lower frequencies; those representing the **base** process relatively higher frequencies.
- Primary and secondary auditory cortex are neighbouring temporal-lobe processing regions.

Tonotopic organisation = an orderly map of sound frequency.
Retinotopic organisation = an orderly map of visual space.

10 Binaural processing: where is the sound?

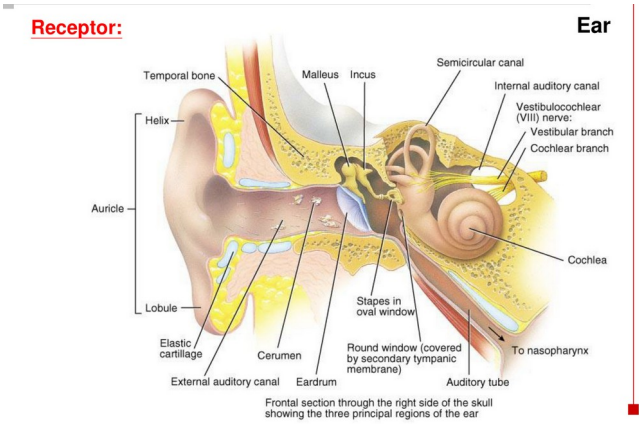
Sound reaches the two ears → auditory signals compared
→ **superior olivary complex detects inter-aural differences**
→ binaural sound processing → **lateralisation**



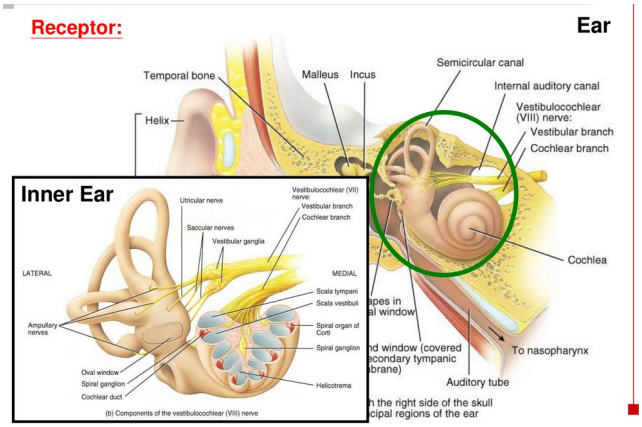
Cue	What is compared?	Example: a source on the left
ITD Inter-aural time difference	Arrival time of sound at the two ears.	Sound reaches the left ear earlier than the right.
IID Inter-aural intensity difference	Sound intensity at the two ears.	Sound is often stronger at the nearer left ear; the head reduces sound reaching the opposite ear.

- "Binaural" means using both ears. "Lateralisation" identifies the side from which a sound comes.
- **ITD and IID help identify location; tonotopy identifies frequency.** These are different features of the same sound.

11 Vestibular apparatus: organs & stimuli



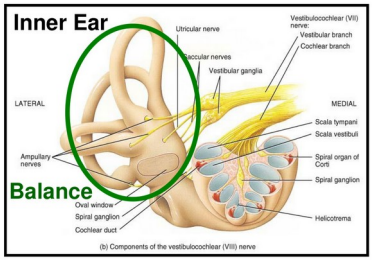
The vestibular receptors are in the inner ear.



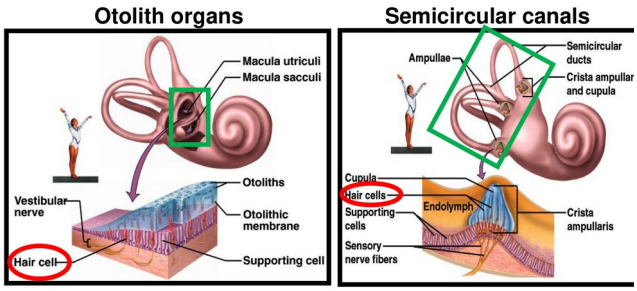
Vestibular and cochlear apparatus lie together.

Vestibular apparatus:

- Otolith organs
  - Utricle
  - Saccule
- 3 Semicircular canals



Two otolith organs + three semicircular canals



Detect head movement

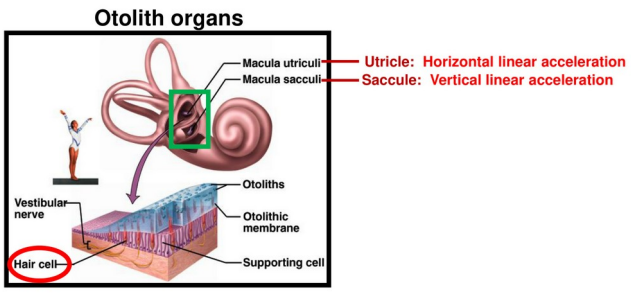
Different mechanical structures detect different movements.

Structure	Main stimulus	Receptor site
Utricle	Horizontal linear acceleration; head tilt.	Macula within an otolith organ.
Saccule	Vertical linear acceleration; head tilt.	Macula within an otolith organ.
Three semicircular canals	Angular acceleration / changes in head rotation.	Crista ampullaris within an ampulla.

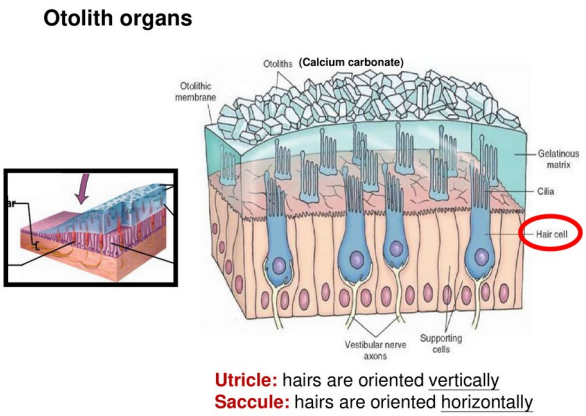
Head movement → hair-cell deflection
→ vestibular branch of CN VIII → brainstem vestibular processing

- Both systems use mechanoreceptor hair cells; their surrounding structures determine which movement bends the bundles.

12 Otolith organs: macula, gel & crystals



Utricle and saccule detect linear acceleration.



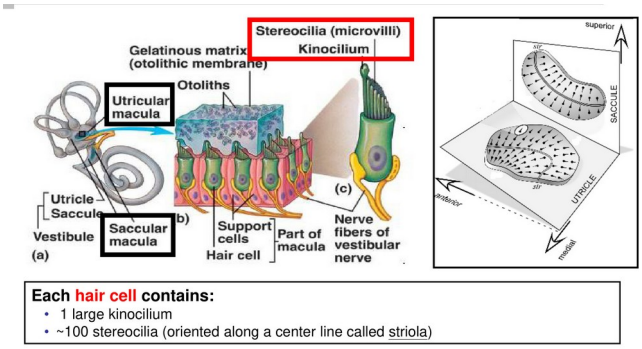
Calcium carbonate crystals add mass to the membrane.

Component	Structure → mechanical role
Macula	Sensory patch containing vestibular hair cells and supporting cells.
Otolithic membrane	Gelatinous matrix over the hair cells; hair bundles project into it.
Otoliths / otoconia	Calcium carbonate crystals on the matrix → add mass and make it responsive to inertia and gravity.
Utricular orientation	Macula is mainly horizontal in an upright head; bundles project mainly vertically → sensitive to **horizontal shear / linear acceleration**.
Saccular orientation	Macula is mainly vertical; bundles project mainly horizontally → sensitive to **vertical shear / linear acceleration**.

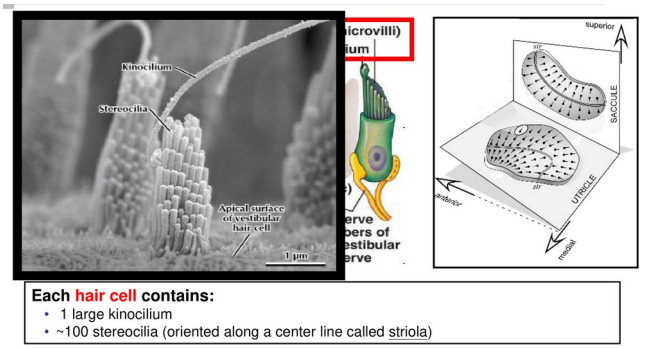
Hair-bundle orientation is not the same as the acceleration detected.
Movement of the overlying membrane bends the bundle sideways.

- The utricle and saccule also indicate head orientation relative to gravity.

13 Vestibular hair cells: direction changes firing



Hair bundles and macular orientation



A kinocilium lies beside the stereocilia.

- A vestibular hair cell has **one large kinocilium and approximately 100 stereocilia**. Stereocilia are specialised microvillus-like projections.
- The **striola** is a curved reference line across the macula. Hair-cell polarities vary across it, allowing the organ to encode different directions.

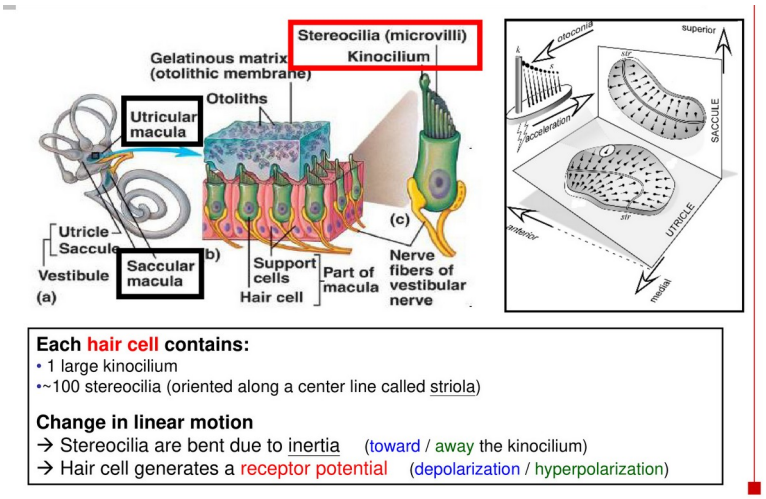
Bundle deflection	Hair-cell response	CN VIII vestibular afferent
Toward the kinocilium	Mechanotransduction-channel opening ↑ → inward current ↑ → **depolarisation** → transmitter release ↑	**Firing rate ↑**
Away from the kinocilium	Channel opening ↓ → inward current ↓ → **hyperpolarisation** → transmitter release ↓	**Firing rate ↓**
Resting position	Intermediate receptor activity and tonic transmitter release.	Baseline firing allows increases AND decreases to signal direction.

Toward → depolarise → fire more.
Away → hyperpolarise → fire less.

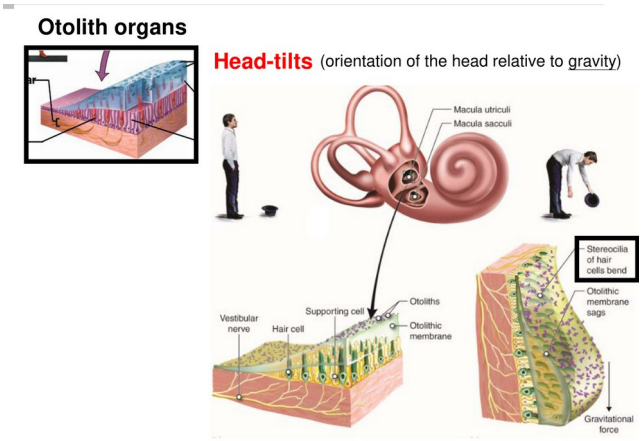
- The receptor potential is produced by the hair cell; action-potential frequency changes in the associated vestibular nerve fibre.

14 Otolith mechanics: acceleration & tilt

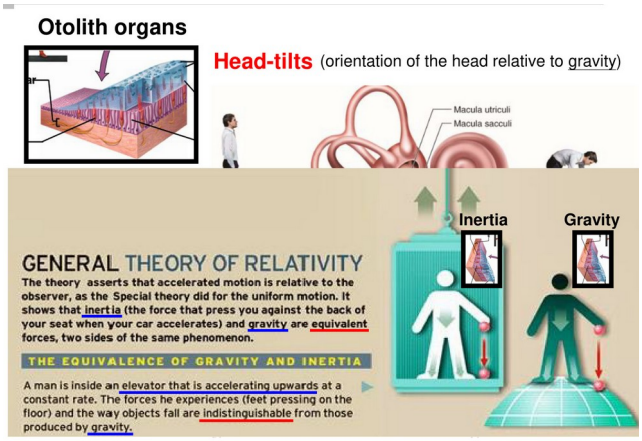
Linear acceleration: head and macula move
→ inertia makes the heavy otolithic membrane lag relative to them
→ shearing force bends hair bundles toward / away from the kinocilium
→ receptor potential changes → **vestibular afferent firing changes**



Head tilt: direction of gravity relative to the macula changes
→ weighted otolithic membrane shifts → sustained bundle deflection
→ information about **head orientation relative to gravity**



Gravity shifts the weighted membrane when the head tilts.

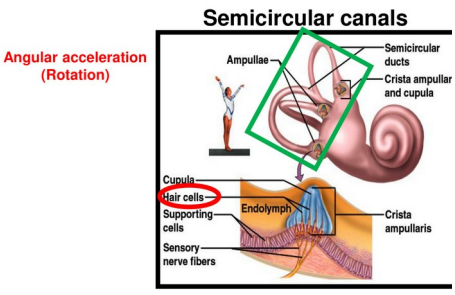


Gravity and linear acceleration can produce similar forces.

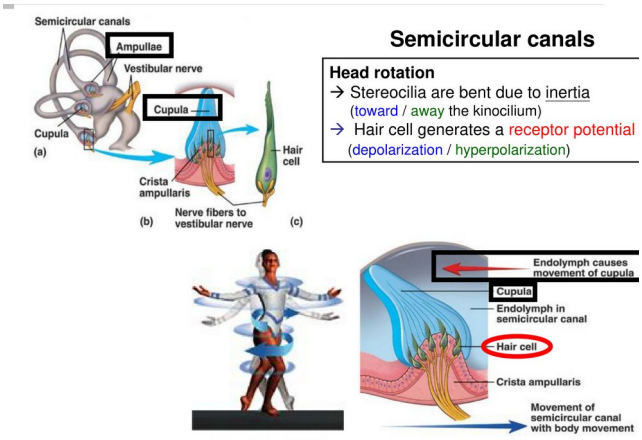
- **Inertia** describes resistance to a change in motion; the relative lag creates shear during acceleration.
- **Gravity** can create shear during a maintained tilt even without ongoing translation.
- Otolith signals alone can be ambiguous between tilt and translation. The brain combines them with canal, visual and body-position information.

15 Semicircular canals: rotation & inertia

Head angular acceleration → canal moves with the skull
→ endolymph initially lags because of inertia
→ cupula deflects → stereocilia bend toward / away from kinocilium
→ hair-cell depolarisation / hyperpolarisation
→ vestibular CN VIII firing increases / decreases



Semicircular canals detect angular acceleration.



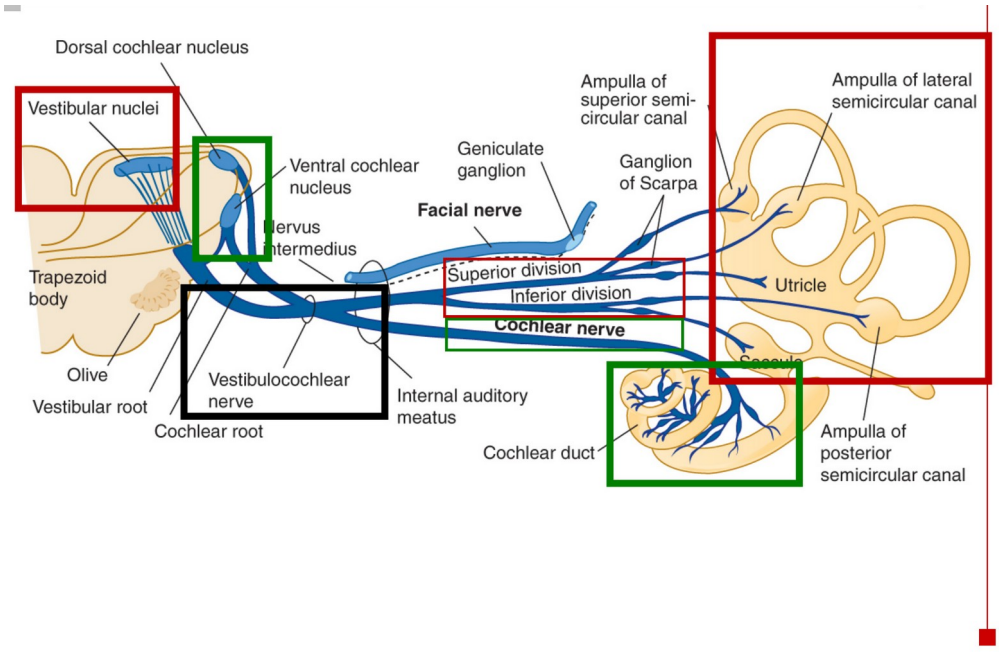
Relative fluid movement bends the cupula.

Structure	Key relationship
Three canals	Anterior / superior, posterior and lateral canals occupy different planes → detect rotation in different directions.
Ampulla	Expanded region of a semicircular duct containing the sensory apparatus.
Crista ampullaris	Sensory ridge containing hair cells and supporting cells.
Cupula	Gelatinous structure covering the hair bundles; deflected by relative endolymph movement.

- The **cupula** is the canal's mechanical coupling structure; the **otolith-loaded membrane** belongs to the utricle and saccule.
- During prolonged constant-speed rotation, the fluid increasingly moves with the canal and the response wanes. A new change in rotational motion renews relative displacement.

16 Vestibular nerve & central functions

Hair cells in **utricle / saccule / semicircular canals**
→ vestibular afferents (cell bodies in **Scarpa's vestibular ganglion**)
→ vestibular branch of **CN VIII** → vestibular nuclei in brainstem
→ connections for perception, postural control and eye stabilisation



Major function	Central connection / outcome
Subjective motion & spatial orientation	Ascending pathways support conscious awareness of movement and position.
Muscle activity & posture	Vestibular connections to spinal motor circuits and the cerebellum adjust posture and help prevent falling.
Eye stabilisation	Connections to ocular motor nuclei generate the vestibulo-ocular reflex (VOR).

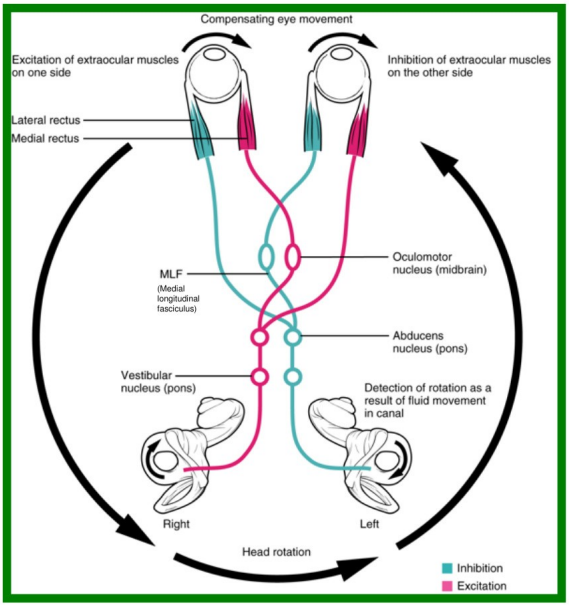
- The vestibular nerve has superior and inferior divisions serving different parts of the vestibular apparatus.
- The cochlear branch instead carries hearing signals from the cochlea to the dorsal / ventral cochlear nuclei.
- Both branches travel together as the **vestibulocochlear nerve (VIII)** through the internal auditory meatus. The nearby facial nerve is a separate cranial nerve (VII).

17 Vestibulo-ocular reflex: keep the image stable

Head turns in one direction → vestibular input changes
→ vestibular nuclei → ocular motor nuclei
→ compensating eye movement in the OPPOSITE direction
→ gaze remains stabilised on the target

Major functions of vestibular sensation:

- **Subjective perception of motion & spatial orientation**
- **Adjustment of muscle activity & posture (to prevent falling)**
- **Stabilization of eyes in space during head movements**



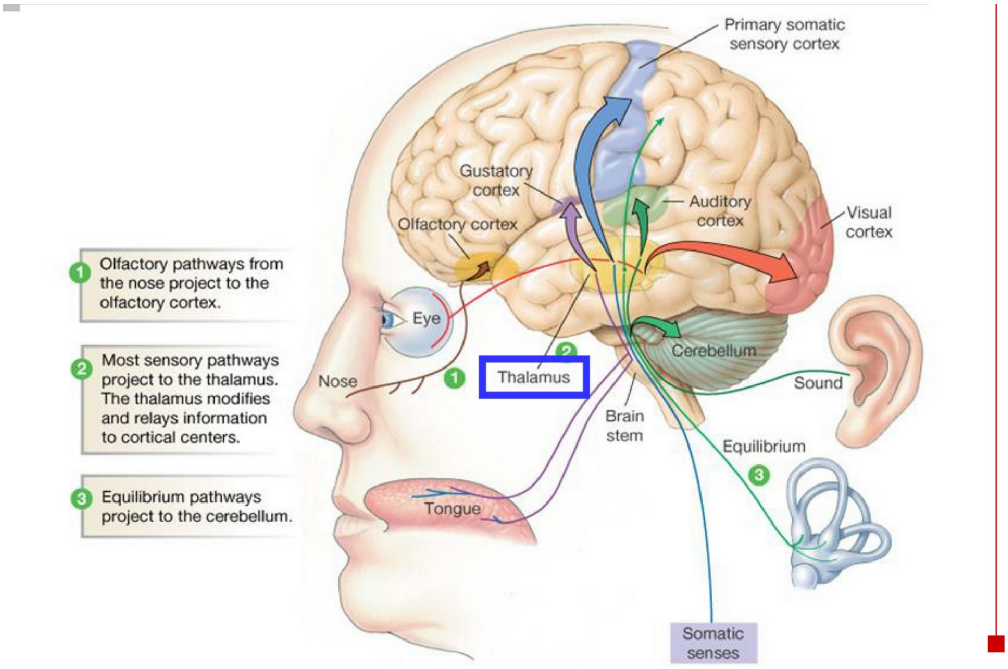
Vestibulo-ocular reflex (VOR)

Horizontal example: turn the head to the right

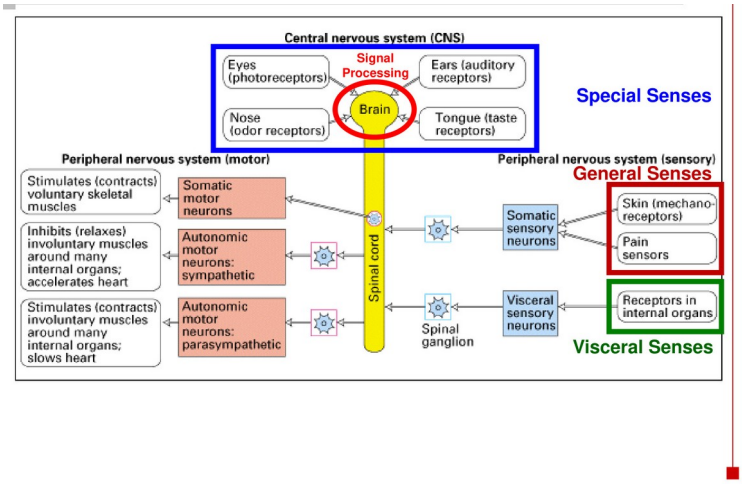
Step	Coordinated response
Rightward head turn	Right horizontal-canal afferent activity ↑; left activity ↓.
Brainstem connection	Right vestibular output activates the left abducens pathway. Interneuronal connections through the **MLF** coordinate the opposite oculomotor nucleus.
Muscles that contract	**Left lateral rectus (CN VI) + right medial rectus (CN III)** → both eyes turn left.
Antagonist muscles	Right lateral rectus and left medial rectus activity decrease to permit the compensating movement.

MLF = medial longitudinal fasciculus. Abducens nucleus: pons. Oculomotor nucleus: midbrain. The horizontal circuit uses CN VI and CN III to coordinate the two eyes.

18 Sensory signals have several central destinations



Pathway	Main relationship
Most ascending sensory pathways	Relay through the **thalamus** before reaching their respective cortical regions.
Olfaction	Reaches primary olfactory cortex without an obligatory initial thalamic relay.
Vestibular / equilibrium pathways	Reach brainstem and **cerebellar** circuits for coordination, as well as pathways for perception and eye movement.



- **General senses:** signals from skin, muscles and body structures. **Visceral senses:** signals from internal organs. **Special senses:** specialised receptor systems such as the eye and ear.
- Sensory processing can guide somatic motor output to skeletal muscle and autonomic motor output to visceral effectors.

19 Sensation, perception & awareness

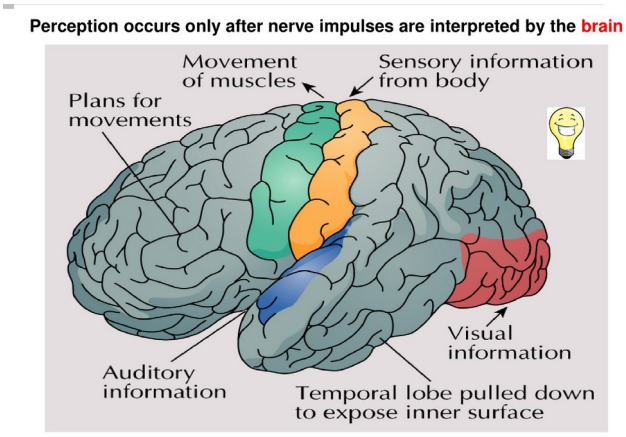
Sensation	Perception
Detect a stimulus and transmit its signal to the brain.	**Interpret the sensory input and recognise what is sensed.**
Begins at a receptor and its sensory pathway.	Depends on central processing and is affected by experience.

Stimulus → receptor response → action potentials in sensory afferents
→ impulses reach the brain → signals translated into information
→ information interpreted → **perception / awareness**

Steps involved in sensory perception:

1. Stimulus is **detected** by the receptor (leading to the generation of action potentials)
2. Action potentials are **conducted** to the brain
3. Action potentials are **translated** into information
4. Information are **interpreted**
5. **Perception (awareness)** of the stimulus

Detection and transmission precede interpretation.



Cortical regions process different types of information.

- **Perception occurs only after the brain interprets the nerve impulses.**
- For hearing and balance, the hair cell generates a receptor potential and releases transmitter; the sensory nerve then carries action potentials.
- The same input can also support automatic actions, such as postural adjustment or VOR, without waiting for a conscious decision.

20 Integrated hearing & balance revision

Feature	Audition	Otolith organs	Semicircular canals
Stimulus	Sound vibration	Linear acceleration / head tilt	Angular acceleration
Receptor site	**Organ of Corti**	**Macula**	**Crista in ampulla**
Mechanical coupling	Basilar motion / tectorial shear	**Gelatinous membrane + otoliths**	**Cupula + endolymph motion**
Hair-cell response	Excitatory bundle bend → K^+ influx → depolarisation	Toward kinocilium → depolarisation; away → hyperpolarisation	Toward kinocilium → depolarisation; away → hyperpolarisation
Nerve branch	Cochlear / auditory VIII	Vestibular VIII	Vestibular VIII
Central outcome	Frequency, loudness and sound location	Orientation, posture and movement perception	Rotational sensation, posture and VOR

Key arrows to retain

- **Eardrum → malleus → incus → stapes → oval window → cochlear fluid → basilar membrane.**
- **Hair-bundle bend → K^+ enters → depolarisation → Ca^{2+} enters → transmitter release.**
- **Cochlear base: narrow / stiff → high frequency. Apex: wide / floppy → low frequency.**
- **Superior olive: ITD + IID → binaural lateralisation. MGN → temporal auditory cortex.**
- **Utricle → horizontal acceleration. Saccule → vertical acceleration. Canals → angular acceleration.**
- **Vestibular bundle toward kinocilium → firing ↑; away → firing ↓.**
- **VOR: head right → eyes left; head left → eyes right.**

